

Classifying Educational Materials by Grade Level

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1 Introduction

In February of 1975, a group of researchers at the Naval Technical Training Command released a report entitled "Derivation of New Readability Formulas (Automated Readability Index, Fog Count and Flesch Reading Ease formula) for Navy Enlisted Personnel". This 51-page report provided guidance for using metrics such as Flesch Reading Ease (FRE), developed in 1948 by Austrian author Rudolph Flesch, to gauge how difficult a text would be to read. The motivation behind this study was a directive to ensure that the average Naval enlisted would be able to read and understand technical documents for equipment.

The FRE formula is still considered a valid metric today. The state of Connecticut's Plain Language Act is a bill that requires any insurance policy text must score a 45 or higher on the FRE in order to be considered readable.

The purpose of this project is to determine whether a Multinomial Naive Bayes Classifier or Logistic Regression classifier can be adapted to a dataset using FRE features, in order to accurately classify the grade-level of educational text resources. FRE has drawbacks such as its dependency on text length and its inability to examine sentence structure. This project will investigate FRE characteristics of texts as features in modern classifiers.

2 Dataset

The dataset used in this experiment is the ScienceQA dataset (Pan Lu, et.al, 2022), a collection of text material from US grades kindergarten to 12. Each entry in the dataset includes 13 arbitrary features: image, question, choices, answer, hint, task, grade, subject, topic, category, skill, solution. In the 'grade' feature, each instance is manually annotated with its intended grade level. The dataset is predivided into a train/dev/test split of 12,726,

4,241, and 4,241 instances respectively.

The structure of each instance before pruning is as follows:

- image : Contextual image
- question : Prompt relating to the lecture
- choices : Multiple choice answer with 1 correct to the question
- answer : Index of choices corresponding to the correct answer
- hint : Hint to help answer the question
- task : Task description
- grade : Grade level from K-12
- subject : High level
- topic : natural-sciences, social-science, or language-science
- category : A subcategory of topic
- skill : A description of the task required
- lecture : A relevant lecture that a question is generated from
- solution : Instructions on how to solve the question

2.1 Preprocessing

In preprocessing, I removed the image feature from each instance as the included image is irrelevant to the task of text-classification. Additionally, I removed 'choices', 'hint', 'answer', 'task', 'subject', 'topic', 'category', and 'skill' as they either contributed nothing to classification (i.e. multiple-choice answers) or may have caused a skew in the results based on subject or required skill.

The following is an example of a pruned instance:

- question : 'Which of these states is farthest north?'
- grade : 'grade2'
- lecture : 'Maps have four cardinal directions, or main directions. Those directions are north, south, east, and west. A compass rose is a set of arrows that point to the cardinal directions. A compass rose usually shows only the first letter of each cardinal direction. The north arrow points to the North Pole. On most maps, north is at the top of the map.'
- 'solution': 'To find the answer, look at the compass rose. Look at which way the north arrow is pointing. West Virginia is farthest north.'

2.2 N-gram Features

The remaining text fields 'lecture' and 'question' were tokenized and converted into n-gram dictionaries using NLTK's tokenizer and n-gram functions, counted, and passed as a dictionary of n-gram:counts into the classifiers.

2.3 Syllable Features

After parsing text fields from each instance into tokens, I ran a loop for each token through the NLTK cmudict module which returned the count of phonemes in each token, if that token has an entry in cmudict. In cmudict, each consonantal phoneme ends in a character and each vowel phoneme in a digit. By checking the ending of each phoneme code, a counter of each vowel phoneme kept track of an estimated number of syllables in each word. Dividing the total number of syllables by the total number of words returned an average syllable-per-word feature for each instance in the dataset.

3 Dev Set Results

3.1 Multinomial Naive Bayes

The MNB classifier had a baseline accuracy of 48.9 for unigrams, 54.1 for bigrams, and 57.2 for trigrams.

Grid search tuning found that an alpha value of 0.1 returned the highest accuracy for all three feature sets.

Phonemes, on the other hand, returned significantly poorer results. The MNB classifier resulted in a baseline accuracy of 16.77 using type-text ratio

alpha	acc	alpha	acc
0.1	0.5253	0.6	0.5076
0.2	0.5232	0.7	0.5088
0.3	0.5163	0.8	0.5062
0.4	0.5112	0.9	0.5027
0.5	0.5097	1.0	0.5005

Table 1: Grid search on MNB Unigrams

alpha	acc	alpha	acc
0.1	0.5713	0.6	0.5578
0.2	0.5628	0.7	0.5562
0.3	0.5595	0.8	0.5576
0.4	0.5567	0.9	0.5564
0.5	0.5543	1.0	0.5562

Table 2: Grid search on MNB Bigrams

alpha	acc	alpha	acc
0.1	0.5729	0.6	0.5649
0.2	0.5717	0.7	0.5642
0.3	0.5703	0.8	0.5673
0.4	0.5666	0.9	0.5696
0.5	0.5659	1.0	0.5706

Table 3: Grid search on MNB Trigrams

and average syllables per word as features. Performing a grid search resulted in an accuracy of 17.38 for all alpha values.

3.2 Logistic Regression

I initially found that the default value of 100 iterations was not enough for model convergence. After testing, I found that 10000 to 12500 iterations was enough to allow the model to converge. The baseline accuracy for the LR classifier for unigrams was 73.98, for bigrams 78.04, and trigrams 78.26.

The grid search consisted of testing over five different solvers ('lbfgs', 'newton-cg', 'liblinear', 'sag', 'saga') and 6 different C parameters (1, 5, 10, 25, 50, 100).

The highest overall results for LR-Unigrams came with the lbfgs solver, and a C value of 25.

C-val	acc	C-val	acc
1	0.7955	25	0.8120
5	0.8094	50	0.8118
10	0.8106	100	0.8113

Table 4: Grid search on LR unigrams, lbfgs solver.

For bigram features, grid search found liblinear

with a C value of 5 to yield the highest accuracy scores.

C-val	acc	C-val	acc
1	0.809	25	0.8137
5	0.8144	50	0.8137
10	0.8141	100	0.8134

Table 5: Grid search on LR bigrams, liblinear solver

For trigram features, the lbfgs solver yielded the highest scores at C values 5 and 25.

C-val	acc	C-val	acc
1	0.8130	25	0.8141
5	0.8141	50	0.8134
10	0.8137	100	0.8139

Table 6: Grid search on LR trigrams, lbfgs solver

When performing the same experiments using the syllable data instead of n-gram features, the LR classifier achieved a baseline accuracy of 14.03. Grid search tuning yielded the highest accuracy scores on the liblinear solver with a C-value of 1.

C-val	acc	C-val	acc
1	0.2199	25	0.2126
5	0.2145	50	0.2126
10	0.2122	100	0.2124

Table 7: Grid search on LR syllables, liblinear solver

4 Test Set Results

This section contains the test results for both classifiers.

4.1 Multinomial Naive Bayes

The test set yielded the highest scores for trigram features with an alpha of 0.1. The test set on syllable features yielded an overall accuracy of 0.16, and the only label that received any score was the grade4 label with a precision of 0.16, recall of 1.00, and f1-score of 0.27. Investigating the data, inspection of the feature building process, and the consistency of poor results with the syllable features lead me to believe that if there is an issue with the data here, it is beyond the scope of this project.

4.2 Logistic Regression

The test results for LR for each tuned hyperparameter showed little variation, however the unigram

	precision	recall	F1	Support
accuracy			0.54	4241
macro avg	0.57	0.62	0.56	4241
weighted avg	0.58	0.54	0.54	4241

Table 8: Test results, MultinomialNB unigrams, alpha 0.1

	precision	recall	F1	Support
accuracy			0.56	4241
macro avg	0.57	0.63	0.57	4241
weighted avg	0.58	0.56	0.55	4241

Table 9: Test results, MultinomialNB bigrams, alpha 0.1

	precision	recall	F1	Support
accuracy			0.56	4241
macro avg	0.59	0.63	0.58	4241
weighted avg	0.58	0.56	0.55	4241

Table 10: Test results, MultinomialNB trigrams, alpha 0.1

	precision	recall	F1	Support
accuracy			0.016	4241
macro avg	0.01	0.05	0.02	4241
weighted avg	0.03	0.16	0.04	4241

Table 11: Test results, MultinomialNB syllables, alpha 0.1

feature set did have the highest overall scores on the test set. The syllable set, as expected, returned much lower scores, but did perform better overall than the MultinomialNB classifier.

	precision	recall	F1	Support
accuracy			0.79	4241
macro avg	0.82	0.83	0.82	4241
weighted avg	0.79	0.79	0.79	4241

Table 12: Test results, LR Unigram, C = 25

	precision	recall	F1	Support
accuracy			0.73	4241
macro avg	0.78	0.76	0.76	4241
weighted avg	0.74	0.73	0.73	4241

Table 13: Test results, LR Bigram, C = 5

	precision	recall	F1	Support
accuracy			0.73	4241
macro avg	0.78	0.75	0.76	4241
weighted avg	0.74	0.73	0.73	4241

Table 14: Test results, LR Trigram, C = 5

	precision	recall	F1	Support
accuracy			0.20	4241
macro avg	0.10	0.12	0.10	4241
weighted avg	0.16	0.20	0.16	4241

Table 15: Test results, LR Syllables, C = 1

5 Discussion

The abysmal results of syllable counts and type-text ratio as features are likely due to the fact that those features simply do not carry enough information to correlate with grade-level classification. Out of the box, both classifiers performed better on n-gram feature sets. The MultinomialNB model achieved accuracy scores between 55 and 60 for n-gram feature sets, which is not terrible. There is some good information here, and further testing could include doing counts for individual phonemes present. Additionally, with more time I would be able to get pyphen working and investigate whether the presence of particular morphemes contributes to higher text complexity. As Rudolph Flesch wrote in 'How to Write Plain English', removing affixes is the most effective way to simplify writing.

6 Conclusion

With this investigation I intended to find some relevance for FRE metrics in text classifiers. Since the experiment was limited to two classifiers, it is very possible that neither of these were optimal for what I was trying to test. At the start, I suspected that there would be some correlation between syllable density and text complexity. The results of this investigation demonstrated that my suspicion was wrong, or at the very least, misguided.